Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA International Joint R&D Program*

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	***Name***	*Dr. Daniel Siegmund*	***Email***	*Daniel.siegmund@umsicht.fraunhofer.de*
Institute Name	*Fraunhofer Institute for Environmental, Safety, and Energy Technology UMSICHT*			
Address	*Osterfelderstr. 3, 46047 Oberhausen, Germany*			

2. Technology Overview

Title of Proposed Technology	*Electro-Catalytic Hydrogenation of Liquid Organic Hydrogen Carriers Using Low-Palladium and Palladium-Free Catalyst Systems*			
Korea's 12 National Strategic Technology Fields	*Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )*			
Industrial Technology Classification 1)	***Technology Field 1** (Large Category)*		***Technology Field 2** (Large Category)*	
	Energy and Resources		Chemicals	
	Medium Category	Medium Category	Medium Category	Small Category
	New & Renewable Energy	Hydrogen	Chemical Processes	Basic Organic Material Process Technologies
Concept of Proposed Technology	Liquid Organic Hydrogen Carriers (LOHCs) represent a highly attractive solution for large-scale hydrogen storage and transportation due to their liquid nature, compatibility with existing infrastructure, and high volumetric hydrogen density. However, current LOHC technologies rely heavily on **palladium-based catalysts and thermally driven hydrogenation**, resulting in high capital and operational costs. The proposed technology focuses on the **development of advanced hydrogenation catalyst systems with reduced or eliminated palladium content**, combined with **electro-catalytic and electro-assisted hydrogenation concepts**. By integrating electrochemical know-how from Fraunhofer UMSICHT with catalyst development and manufacturing expertise from Korean SMEs, the project aims to significantly lower hydrogenation costs while maintaining high activity, selectivity, and long-term stability.			

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	The technology will be applicable to multiple LOHC systems (e.g., aromatic and heterocyclic carriers) and designed to support Korea's emerging hydrogen transport and distribution network. Beyond LOHC hydrogenation, the underlying concept represents a broader **electro-organic technology platform**, which can be further advanced toward the electrochemical design and fabrication of tailored organic molecules. Such electrochemical routes enable the development of specialty molecules with adjustable structures and properties, offering additional flexibility and value creation for hydrogen storage, transport, and related chemical applications. Describe the core concept, main functions, and potential uses of the proposed technology
Rationale for Joint R&D Project	Korea is rapidly expanding its hydrogen economy and requires **safe, scalable, and cost-effective hydrogen storage technologies** to support national hydrogen infrastructure. LOHC systems are particularly well-suited for Korea due to their compatibility with existing logistics, chemical handling, and industrial distribution networks. However, reducing reliance on noble metals and lowering hydrogenation energy demand are key barriers to commercialization. Addressing these challenges requires: • Advanced electrochemical system design and process integration (Fraunhofer UMSICHT) • Catalyst materia and storing moleculel development, scaling, and manufacturability (Korean SMEs) A joint R&D project enables fast validation under realistic conditions, reduces technical risk for SMEs, and accelerates the transition from laboratory-scale concepts toward industrial implementation
Key Words	1. Hydrogen Storage and Transport 2. Liquid Organic Hydrogen Carrier 3. Low PGM-loaded catalyst 4. Electro-Organic Synthesis 5. Electrolysis

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (4)
R&D Objectives	Develop hydrogenation catalysts for LOHC systems with **significantly reduced palladium content or palladium-free compositions** Integrate electrochemical hydrogen activation and hydrogenation concepts to reduce overall energy demand Demonstrate stable LOHC hydrogenation performance under industrially relevant conditions Generate performance and durability data suitable for SME-driven scale-up decisions			
R&D Target Specifications	Palladium reduction: ≥50% compared to conventional Pd-based LOHC catalysts Hydrogenation selectivity: ≥95% for target LOHC compounds Catalyst stability: ≥100 hours continuous operation without significant deactivation Energy consumption: measurable reduction compared to purely thermal hydrogenation benchmarks			
R&D Plan	The joint R&D project is structured in two consecutive phases in accordance with the TIPA International Joint R&D Program: a **6-month pre-study (Phase 1)** followed by a **Main R&D Phase of approximately 2.5 years (Phase 2)**. The Phase 1 activities focus on rapid feasibility validation using accelerated methodologies, while Phase 2 targets technology maturation, robustness, and application expansion.			
	Phase 1: Pre-study (6 months) **Objective:** Rapid technical validation, catalyst down-selection, and definition of an optimized electro-catalytic LOHC hydrogenation concept using accelerated and data-driven methods. **Task 1: Selection of LOHC Systems and Definition of Target Specifications** **Timeline:** Months 1–2 • Selection of representative liquid organic hydrogen carriers relevant for hydrogen storage and transport within the Korean hydrogen infrastructure • Definition of hydrogenation and dehydrogenation targets, including hydrogen capacity, selectivity, energy demand, and cycling requirements • Establishment of benchmark data using conventional thermally driven, palladium-based catalyst systems **Milestones:** • LOHC candidates selected • Target specifications and benchmark KPIs defined **Task 2: Accelerated Catalyst Screening Using Advanced Electrochemical Platforms** **Timeline:** Months 1–4 • Deployment of accelerated electrochemical screening stations enabling parallel testing of multiple catalyst formulations and operating conditions • High-throughput evaluation of low-palladium and palladium-free catalyst systems with respect to activity, selectivity, and hydrogen uptake efficiency • Rapid identification of promising catalyst compositions and process windows under electro-catalytic and electro-assisted conditions **Milestones:** • Shortlist of high-performance catalyst systems • Quantified performance advantage over benchmark Pd-based catalysts			

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Task 3: Data-Driven Parametric Optimization Using Bayesian and AI-Based Methods
Timeline: Months 3–6

- Systematic parametric optimization of electro-catalytic LOHC hydrogenation using Bayesian optimization and AI-supported design-of-experiments
- Optimization of key parameters including catalyst composition, current density, temperature, pressure, electrolyte composition, and substrate concentration
- Identification of optimal operating conditions maximizing selectivity, hydrogenation rate, and catalyst utilization

Milestones:

- Optimized parameter sets for selected LOHC–catalyst combinations
- Demonstrated improvement in selectivity and catalytic activity

Task 4: Feasibility of Electro-Catalytic LOHC Cycling
Timeline: Months 4–6

- Initial evaluation of electrochemical hydrogenation–(thermal) dehydrogenation cycling using pure LOHC substrates
- Assessment of catalyst stability and performance retention over repeated charging and discharging cycles
- Identification of early degradation mechanisms relevant for long-term operation

Milestones:

- Proof-of-concept demonstration of reversible LOHC cycling
- Defined cycling-related risks and requirements for Phase 2

Outcome of Phase 1:

- Down-selected LOHC systems and catalyst candidates
- Validated electro-catalytic hydrogenation concept supported by accelerated and AI-optimized data
- Clearly defined technical scope, risks, and go/no-go criteria for Phase 2

Phase 2: Main R&D Phase (~2.5 years)
Objective: Technology maturation, cycling robustness, reaction scope expansion, and preparation for industrial implementation.

Task 5: Advanced Catalyst Optimization and Scale-Relevant Synthesis
Timeline: Months 1–12 (Phase 2)

- Further optimization of catalyst composition and structure based on Phase 1 results
- Development of scalable synthesis and processing routes in collaboration with Korean SMEs
- Validation of reproducibility and manufacturability

Milestones:

- Optimized, scale-relevant catalyst formulations

Task 6: Electro-Catalytic Reactor and Process Development
Timeline: Months 6–18 (Phase 2)

- Design and optimization of electro-catalytic and hybrid electro-thermal reactor concepts
- Integration of optimized catalysts into reactor systems suitable for LOHC processing
- Energy efficiency and operational stability optimization

Milestones:

- Validated reactor and process configuration

Task 7: Long-Term LOHC Cycling and Durability Evaluation
Timeline: Months 12–24 (Phase 2)

- Extended hydrogenation–dehydrogenation cycling using pure LOHC substrates
- Quantitative assessment of performance degradation, catalyst lifetime, and regeneration strategies
- Optimization of operating strategies to maximize cycle efficiency and catalyst durability

	Milestones: • Demonstrated stable multi-cycle LOHC operation • Defined lifetime and maintenance indicators
	Task 8: Expansion Toward Electro-Organic Hydrogenation and Alternative Reactions **Timeline:** Months 18–30 (Phase 2) • Exploration of alternative electro-organic hydrogenation reactions beyond LOHC systems using the developed catalyst platform • Evaluation of catalyst versatility for hydrogenation of other organic substrates relevant to specialty chemicals • Identification of additional application fields for electro-organic synthesis **Milestones:** • Demonstrated catalyst applicability beyond LOHC hydrogenation • Identified SME-driven opportunities for product and market diversification
	Final Project Outcomes The project will deliver a validated electro-catalytic LOHC hydrogenation technology platform characterized by accelerated development, AI-driven optimization, reversible cycling capability, and extended applicability toward electro-organic synthesis. This enables Korean SMEs to expand from hydrogen production into hydrogen storage, transport, and advanced electro-organic chemical manufacturing.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer UMSICHT will act as the **technical lead and system integrator** of the joint R&D project, providing electrochemical, catalytic, and process-engineering expertise to support the development of advanced LOHC hydrogenation and electro-organic synthesis technologies. **Core Technical Contributions** • Development and operation of electro-catalytic and electro-assisted hydrogenation concepts for Liquid Organic Hydrogen Carriers (LOHCs) • Design, implementation, and operation of accelerated electrochemical screening platforms enabling rapid evaluation of catalyst systems and operating conditions • Application of data-driven optimization approaches, including Bayesian optimization and AI-assisted experimental design, to maximize catalyst activity, selectivity, and energy efficiency **Research Scope and Methodology** • Selection and benchmarking of LOHC systems and reference catalyst technologies • Electrochemical testing, performance evaluation, and durability assessment under industrially relevant conditions • Investigation of reversible hydrogenation–dehydrogenation cycling behavior and identification of degradation mechanisms • Extension of developed catalyst and process concepts toward electro-organic hydrogenation and alternative electro-organic synthesis reactions **Resources and Infrastructure to be provided** • Access to state-of-the-art electrochemical laboratories, accelerated screening stations, and analytical infrastructure • Provision of experimental protocols, performance benchmarking frameworks, and data evaluation tools • Support for techno-economic assessment and scalability evaluation of catalyst and process concepts **Knowledge Transfer** • Technical coordination of joint R&D activities and alignment of research tasks across consortium partners • Continuous knowledge transfer to Korean SME partners through joint experiments, data sharing, and technical workshops • Support in defining scale-up strategies and follow-up pilot projects for commercialization Through these activities, the research institute will ensure rapid technology development, risk reduction, and effective transfer of electro-organic hydrogen storage and synthesis technologies to Korean SMEs.
Expected R&R of Korean SME Partner	The Korean SME partner is expected to contribute industrial know-how, development capabilities, and application-oriented perspectives relevant to hydrogen production, storage, and transport technologies. Depending on its core competencies, the Korean SME partner may act in one or more of the following roles: • **Electrolysis or cell technology developer:** ○ Provide expertise in electrochemical cell design, materials, and system integration ○ Support the adaptation of electrolysis technologies toward alternative markets beyond hydrogen production, particularly hydrogen storage and transport using LOHC systems ○ Contribute to the development, optimization, and validation of electro-catalytic hydrogenation concepts and reactor configurations • **Chemical or materials manufacturer:** ○ Develop and supply specialized LOHC molecules or tailored organic hydrogen carrier compounds suitable for efficient hydrogenation and dehydrogenation ○ Contribute catalyst materials, supports, or scalable synthesis routes for low-palladium or palladium-free hydrogenation catalysts ○ Support assessment of manufacturability, scalability, and compatibility with existing chemical

	production infrastructure Across all roles, the Korean SME partner will participate in: • Joint definition of technical requirements and target specifications • Experimental validation and performance evaluation from an industrial perspective • Identification of scale-up challenges and commercialization pathways relevant to the Korean hydrogen value chain This collaborative approach enables Korean SMEs to expand their technology portfolio toward hydrogen storage and transport solutions while reducing technical and commercial risks associated with entering the LOHC market.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project will provide Korea with direct access to **advanced electro-organic hydrogen storage technologies**, extending beyond conventional PEM or alkaline electrolysis focused solely on hydrogen production. By addressing **hydrogen storage and transport via Liquid Organic Hydrogen Carriers (LOHCs)**, the project contributes to closing the hydrogen value chain from production to utilization. **Key Technological Impact** • Establishment of electro-catalytic LOHC hydrogenation technologies with **significantly reduced PGM dependency**, lowering material and operational costs • Demonstration of reversible hydrogenation–dehydrogenation cycling, enabling practical hydrogen storage solutions compatible with existing infrastructure • Introduction of **electro-organic synthesis capabilities**, allowing Korea to leverage its strong chemical industry to design and produce tailored organic molecules beyond classical LOHC systems **Commercialization and Industrial Impact** • Creation of new commercialization opportunities through the development of **novel catalysts, electrochemical reactors, and organic hydrogen carrier molecules** • Expansion of Korean SMEs' technology portfolios from hydrogen production into **hydrogen storage, transport, and specialty chemical markets** • Formation of a new technology field at the interface of electrolysis, catalysis, and organic synthesis, enabling product differentiation and export potential **Techno-Economic Impact** • Potential reduction of LOHC hydrogenation costs by **20–40%** through noble-metal reduction and electro-assisted operation • Lower entry barriers for SMEs by reusing existing electrolysis, chemical processing, and manufacturing infrastructure • Long-term contribution to reducing hydrogen logistics costs and strengthening Korea's competitiveness in the global hydrogen economy Overall, the project positions Korea to move beyond hydrogen production technologies toward **integrated hydrogen storage and electro-organic value creation**, combining energy and chemical expertise to unlock high-impact industrial and economic benefits.

6. References

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☐ *Terms:*

1) Industrial Technology Classification

*· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).*

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	*1st Stage: Basic theories and experiments* *2nd Stage: Proof of Concept, such as practical ideas or patents*
	Experimental Stage (3-4)	*3rd Stage: Verification of basic performance at a laboratory scale* *4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale*
	Prototype Stage (5-6)	*5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems* *6th Stage: Production and performance evaluation of prototypes at a pilot scale*
	Productization Stage (7-8)	*7th Stage: Reliability testing and evaluation by potential customers* *8th Stage: Prototype certification and standardization*
	Commercialization Stage (9)	*9th Stage: Commercialization*

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